

Aligned Subgroup Discovery with Partial-Axis Centres: the Price of One Grid and a Certificate That Survives the Search

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Abstract

Subgroup discovery returns a list of conjunctive rules; the rules are read one at a time and, unless the method constrains them, they use different columns and overlap. We study the *aligned* version of the problem: all groups must be cells of one grid spanned by at most four columns, so that they can be shown together in one picture. The constraint has a price. Against free conjunctive rules of the same length at the same budget of conditions, one grid of fully specified cells covers fewer target rows and never more. We show that the price is paid for fixing every axis, not for sharing the columns: allowing a centre to leave some axes of the grid free (a *partial-axis centre*, a bar, plate or layer of the same cube) recovers, under a post-selection certificate, the coverage of free disjoint rules (on the seventeen tables of the study: 2 wins, 6 losses, 40 ties over 48 configuration–budget cells, the losses on `connect_4`; on ten further tables outside it: 2, 0, 77), and closes a median 85 % of the gap under the observed-purity rule (97 % outside the study); against the full grid it is never worse and better in 15 of 48 certified and 41 of 109 observed-purity cells. The candidate partial centres are the cells the exhaustive search over $d \leq 4$ already enumerates, so the family-wise certificate (a union bound over every cell the search can show, proved for rows with heterogeneous success probabilities) is unchanged. We also show that the standard budgeted-coverage rule baseline is not like for like: overlapping rules each of purity $\geq \tau$ form unions whose held-out purity falls below τ in 39–47 % of the splits, and we introduce the disjoint-rule baseline that a method with disjoint groups must be compared against. All numbers are produced by scripts with fixed seeds on 18 tables with pinned hashes.

1 Introduction

A data analyst asks where a value of a categorical target lives. Subgroup discovery [Klösgen, 1996, Wrobel, 1997, Atzmueller, 2015] answers with a ranked list of conjunctions $c_1 = v_1 \wedge \dots \wedge c_\ell = v_\ell$. Two properties of such a list make it hard to read as a whole: the rules use different columns, and they overlap. A reader who wants a picture of the target has to hold each rule separately.

The approach studied here fixes the picture first. A *schema* is a set S of at most four columns; the *grid* on S is the joint partition of the rows by the values of those columns; a *centre* is a cell of purity at least τ with at least m rows. A schema with its centres is one display: a table, a cube, or a cube with a fourth axis stepped through. The search is exhaustive over schemas of every size up to four, so it can find pairs of columns that are jointly informative while neither is alone, which a greedy chain cannot [Krause and Guestrin, 2005]. The software (VSF) shows the winning schema as a lattice with the centres coloured, every scored schema as a point in a coverage landscape, and the cells of the winners as a filtered rule list; this paper is about the search and its guarantees, not the display.

Two questions decide whether this is a method or merely a constraint:

1. **What does one grid cost?** At an equal budget of conditions, how much target coverage does the alignment lose against free rules, and what mechanism loses it?

2. **Is anything shown still true after the search?** The best of $\sum_{d \leq 4} \binom{M}{d}$ schemas was chosen by looking at the target; a purity certificate computed for one schema alone is invalid after that choice.

Contributions.

1. *Partial-axis centres.* We relax the grid constraint by one step: a centre may fix a subset $S' \subset S$ of the schema’s axes. Its candidates are the cells of sub-schemas the search already enumerates; a deterministic disjoint packing keeps coverage a sum and keeps the purity claim for the union. The relaxation removes most of the price of alignment (Section 7) and shows where the price came from: fixing all axes, not sharing the columns (Section 7.1).
2. *A certificate that survives the search.* A cell is certified when its target count reaches $k^*(n, \tau, \alpha/T)$, a heterogeneous-binomial threshold with T the number of distinct row sets among all cells the search can display. The guarantee is a union bound, valid for any rule of schema choice and any dependence between cells, and holds simultaneously for every τ (Section 4). Partial centres do not enlarge T .
3. *An evaluation protocol and a corrected baseline.* Nested cross-validation with the encoding (capacity rule, level merging, family size) fitted on the training fold; held-out purity of the *union* of selected groups; and a disjoint-rule baseline, since budgeted maximum coverage over overlapping rules yields unions below τ in half of the held-out splits (Section 5).

What we do not claim: that one grid covers more than free rules (it does not), or that the shared picture is easier to read (that needs a user study, which we leave to future work).

2 Related work

Subgroup discovery. Subgroup discovery [Klösgen, 1996, Wrobel, 1997, Herrera et al., 2011, Atzmueller, 2015] finds descriptions of row sets with an unusual target distribution. SD-Map [Atzmueller and Puppe, 2006], CN2-SD [Lavrač et al., 2004], beam and best-first search in pysubgroup [Lemmerich and Becker, 2018], diverse top- k (DSSD, van Leeuwen and Knobbe, 2012), MDL-based rule sets and lists [Fischer and Vreeken, 2019, Proença et al., 2022]. Redundancy among top- k subgroups is the problem DSSD and the MDL methods address; the one-grid constraint addresses it structurally, since disjoint cells of one partition cannot overlap. Quality functions are typically size $\times$ lift (WRAcc); the task here, maximum coverage subject to a purity floor, is a different objective, and on *titanic* a WRAcc-ranked list from pysubgroup (Apriori, depth ≤ 4) puts `sex = female` first (purity 0.73) and at $\tau = 0.9$ never reaches τ in any held-out split; with the τ filter applied it is the same candidate family as free rules with a worse selection order (coverage non-monotone in the budget, union purity at τ in 56–64 % of splits at $B \geq 16$), so we compare against the rules baseline directly.

Statistically sound pattern mining. Bonferroni over the pattern space [Webb, 2007], LAMP and Tarone’s trick [Terada et al., 2013, Llinares-López et al., 2015], the two-sample split (exploratory/holdout, Webb, 2007), and the tutorial of Hämmäläinen and Webb [2019] on which corrections are valid after a search. Our certificate is of the Bonferroni family with the family taken over row sets rather than descriptions, with a threshold valid for heterogeneous rows [Hoeffding, 1956]; Tarone’s improvement is available only at a fixed τ and is not used for the interactive setting (Section 4).

Controlled visual exploration. Controlling false discoveries during interactive exploration [Zhao et al., 2017, Zraggen et al., 2018]; visual subgroup discovery in VIKAMINE [Atzmueller and Puppe, 2005]; intersectional subgroup analysis with significance in FairVis [Cabrera et al., 2019]. Displays of cells of a multi-way table with area and colour go back to mosaic plots [Friendly, 1994]. We do not claim novelty for the display; the claim is about the search and the certificate behind it.

Optimal trees and slice finding. A depth-4 decision tree is the most natural competitor for a 4-column description; CART is greedy at the root and loses planted grid structure, while an optimal tree [Lin et al., 2020, Demirović et al., 2022] with the coverage-under-purity objective would be the stronger baseline; the available solvers minimise misclassification plus a leaf penalty, not coverage at a purity floor, and we state the gap as a limitation. Slice Finder [Chung et al., 2019] looks for underperforming slices of a model with the same conjunctive language.

3 Problem and method

3.1 Setting

Rows $i = 1, \dots, N$ with categorical columns $X_{i1}, \dots, X_{iM}$ (numeric columns are binned once, on all rows, and the bins are part of the dataset) and a binary target $z_i \in \{0, 1\}$ with $P = \sum_i z_i$ positives. Purity floor τ above the base rate P/N , size floor m .

Definition 1 (Grid, centre, coverage). *For a schema $S \subseteq \{1, \dots, M\}$, $|S| \leq 4$, the grid $\mathcal{G}(S)$ is the partition of rows by the joint value of the columns in S , after the capacity rule of Section 3.2. A cell c with n_c rows and k_c positives is a centre if $n_c \geq m$ and $k_c/n_c \geq \tau$ (observed-purity rule) or $k_c \geq k^*(n_c, \tau, \alpha/T)$ (certified rule, Section 4). The coverage of S is $\text{cov}(S) = \sum_{c \text{ centre}} k_c/P$.*

The full-grid search returns, for each $d \leq 4$, $S_d^* = \arg \max_{|S|=d} \text{cov}(S)$, ties broken by fewer centres, then smaller mass, then the best per-cell lower bound. Every $\binom{M}{d}$ subset is scored.

3.2 Capacity rule and level merging

A grid may occupy at most $\lfloor N/10 \rfloor$ cells; when it does not fit, the widest column loses its rarest level into an “other” group, one level at a time, until it fits. The rule reads no target and is refitted on the training rows in every fold (Section 5). Levels of an ordered column are merged by adjacency, of a nominal column by frequency; the merge sequence is deterministic (widest column first, ties by column index) and the first state that fits is found by bisection over the step count. The rule bounds the number of cells the display has to draw and the number of hypotheses in the family; it also breaks the nesting of cells across dimensionalities (11.9% of nested pairs on `titanic`), which is why an optimistic-estimate pruning of the search is not attempted here.

3.3 Partial-axis centres

Definition 2 (Partial centre). *A partial centre of schema S is a centre of the grid $\mathcal{G}(S')$ for some non-empty $S' \subseteq S$. It fixes $|S'|$ axes of S and leaves $|S| - |S'|$ free; it costs $|S'|$ conditions. On the display of S it is a bar ($|S'| = |S| - 1$), a plate, or a layer.*

Partial centres of different sub-schemas may overlap. We select a disjoint set by a deterministic greedy: candidates in the order (more positives, fewer conditions, fewer rows, enumeration order), each taken if it shares no row with those already taken. The coverage of S with partial centres is then $\text{cov}^p(S) = \sum_{c \text{ chosen}} k_c/P \geq \text{cov}(S)$, since every full cell is a candidate. The union of disjoint groups each of purity $\geq \tau$ has purity $\geq \tau$.

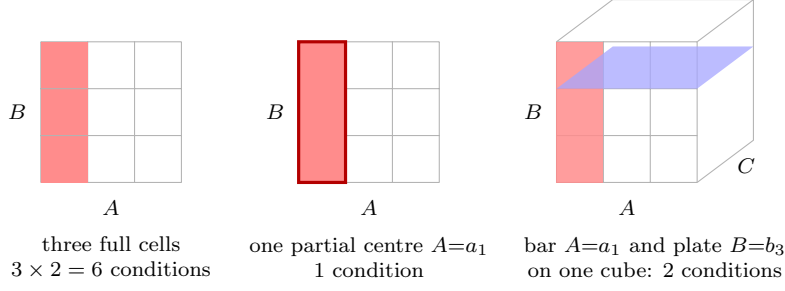


Figure 1: Why fixing every axis costs conditions. Left: on the grid of (A, B) the group $A=a_1$ is three full cells, six conditions. Middle: the same group as one partial centre, one condition; on the display of (A, B) it is a bar. Right: two groups on different columns of one four-column schema are a bar and a plate of the same cube, two conditions; as full cells of a four-column grid they would be $9 + 9$ cells at four conditions each.

Three properties make the relaxation free. (i) The candidates of S are the centres of its sub-schemas, which the exhaustive enumeration visits before S ($d = 1, \dots, 4$ in order); nothing new is computed. (ii) Every candidate is a cell of a partition of some S' with $|S'| \leq 4$, hence a member of the family T of Section 4: the certificate is unchanged. (iii) The display is unchanged: a partial centre is a union of cells of the same cube.

Under a budget of B conditions (Section 5) the packing becomes budgeted maximum coverage over disjoint candidates; we use the cost-effectiveness greedy with lazy evaluation compared against the best single affordable candidate [Khuller et al., 1999], restricted to candidates disjoint from the chosen ones.

4 A certificate that survives the search

Let $\mathcal{T}$ be the set of distinct row sets among the occupied cells of every partition the search can display: the search partition of every schema with $|S| \leq 4$ and, where the capacity rule merged levels, the unmerged partition as well. $T = |\mathcal{T}|$ is fixed once the columns are fixed and does not depend on the target, on τ or on m .

Theorem 1 (Family-wise certificate). *Conditionally on the columns, let the rows be independent with $\Pr(z_i = 1) = p_i$. For a cell c with rows I_c let $\bar{p}_c = \frac{1}{n_c} \sum_{i \in I_c} p_i$. Let*

$$k^*(n, \tau, \lambda) = \min \left\{ k : k \geq \lfloor n\tau \rfloor + 1, \Pr(\text{Bin}(n, \tau) \geq k) \leq \lambda \right\}.$$

Then, with $\lambda = \alpha/T$, the probability that some cell $c \in \mathcal{T}$ with $\bar{p}_c \leq \tau$ has $k_c \geq k^(n_c, \tau, \lambda)$ is at most α , simultaneously over every $\tau \in (0, 1)$ and every rule by which schemas are chosen after seeing the target.*

Proof sketch. k_c is Poisson-binomial with mean $n_c \bar{p}_c$. By Hoeffding's Theorem 4 [Hoeffding, 1956], for $k - 1 \geq n \bar{p}_c$ the upper tail of a Poisson-binomial sum is at most that of $\text{Bin}(n, \bar{p}_c)$; the threshold is defined as the smallest k^* with (i) $k^* \geq \lfloor n\tau \rfloor + 1$ and (ii) $\Pr[\text{Bin}(n, \tau) \geq k^*] \leq \alpha/T$, so that $k^* - 1 \geq n\tau \geq n \bar{p}_c$ and the comparison applies. Condition (i) is not cosmetic: without it the binomial test is anti-conservative for heterogeneous rows ($n = 3$, $\tau = 0.01$, level 0.0298: the binomial threshold is $k = 1$, while rows with $p = (0.03, 0, 0)$, mean 0.01, give $\Pr[k \geq 1] = 0.03$ above the level; by AM–GM the chance of at least one success is smallest when the p_i are equal). Monotonicity of the binomial tail in p gives $\Pr(k_c \geq k^*) \leq \Pr(\text{Bin}(n_c, \tau) \geq k^*) \leq \alpha/T$ for every cell with $\bar{p}_c \leq \tau$. A union bound over the T cells, which are fixed given the columns, gives the family-wise statement regardless of how the displayed schema was selected. Simultaneity in τ follows because the event for a larger τ is contained in that for a smaller one. $\square$

Table 1: Smallest certifiable cell $n_{\min}$ at $\alpha = 0.05$ (family Bonferroni, T distinct row sets over $d \leq 4$). “Pure”: all rows carry the target; “0.85 / 0.95”: observed purity five points above $\tau = 0.8$ / 0.9.

data	N	M	T	pure, $\tau = 0.7$ / 0.9	0.85 / 0.95
synthetic	5000	5	2 100	30 / 102	142 / 521
titanic	891	8	3 102	31 / 105	148 / 542
mushroom	8124	21	49 254	39 / 131	189 / 701
chess	3196	36	303 620	44 / 149	221 / 803
synthetic	5000	30	7 282 600	53 / 179	269 / 984

What the theorem does not say. It bounds the *average* success probability of the rows in a cell, not a population share, and it assumes independence given the columns (families in **titanic** violate it). The guarantee is per (target value, direction, column set). Westfall–Young resampling was considered and rejected: a permutation of the target tests independence, not $\bar{p}_c \leq \tau$, and certifies a cell of purity 0.85 as “> 0.9”. Tarone’s trick removes cells that cannot reach k^* at the given τ but makes the family depend on τ , which breaks simultaneity for an interactive τ ; it is admissible in an experiment with τ fixed in advance and reduces the smallest certifiable cell only logarithmically (on **titanic**, $\tau = 0.9$: $n_{\min} = 105$ under family Bonferroni, 86 with Tarone, Table 1).

Price. Table 1 gives $n_{\min}$, the smallest cell that the family certificate can certify at $\alpha = 0.05$, as a function of the family size T (`experiments/n_min.table.py`). From $M = 5$ to $M = 30$ columns T grows by a factor of 3 500 and $n_{\min}$ by 1.75: the price of the search is logarithmic in the family. The price of purity is not: a cell whose purity is five points above $\tau = 0.9$ needs 520–980 rows at any M , so at $N \leq 500$ the certificate certifies nothing but perfectly pure cells, and at $\tau = 0.9$ nothing on **titanic**.

Power. Power under strict control at $N = 500$ is 1 %, at $N = 1000$ 57 %, at $N = 3000$ 100 % (`experiments/certificate.power.py`: eight 3-level columns, a true centre at purity 0.95, a cell at exactly $\tau = 0.8$, background 0.10; FWER 0 in every setting; the per-schema certificate the product used before has power 0.75 at $N = 500$ but a family-wise false-certificate rate of 68 % on pure noise at $d \leq 4$).

5 Evaluation protocol

Folds. Repeated stratified K -fold (5×5 ; 5×2 for **mushroom** and **adult**, 5×1 for **chess**, **connect.4** and **splice**), identical across methods so that differences are paired. Every fitted quantity — bins are fixed, but the capacity rule, level merging and the family size T — is computed on the training fold; the chosen groups are carried to the held-out rows through the fitted partition (unseen levels go to the “other” group).

Budget. All methods answer the same question: with at most $B \in \{1, 2, 4, 8, 16, 32\}$ conditions, which groups of rows hold the most positives, each at training purity $\geq \tau$ (or certified) and size $\geq m$? A cell of a d -column grid costs d ; a rule of length ℓ costs ℓ ; a tree leaf costs its depth.

Methods. **vsf**: best schema per budget, its top $\lfloor B/d \rfloor$ centres. **vsf_partial**: best schema per budget with partial centres, disjoint greedy. **vsf_greedy**: the nested chain $S_1 \subset S_2 \subset S_3 \subset S_4$ (the greedy the exhaustive search argues against). **rules**: every conjunction of ≤ 4 conditions

occurring in the training rows, budgeted maximum coverage (may overlap). **rules_pure**: **rules** with the union kept at training purity $\geq \tau$. **rules_disjoint**: **rules** with no shared training row. **tree**: CART on one-hot columns, best over depth ≤ 4 , criterion and class weight; qualifying leaves under the same greedy. Under the certified rule the tree’s leaves are not a pre-specified family, so the filter is only a filter, not a certificate. **ssdpp**: SSD++ [Proença et al., 2022], an MDL rule list fitted on the training fold for the two-class target (beam 50, depth ≤ 4 , minimum support m); a rule list is ordered, so its i -th rule applies to the rows no earlier rule matched and its groups are disjoint by construction. Its qualifying rules enter the same budgeted greedy at cost = conditions. This is the strongest published subgroup-set method with a redundancy control; its objective (compression of the whole target column) differs from ours, and the list also holds rules for the other class, which never qualify. The package’s PyPI release needs a compatibility patch for current numpy/pandas that we ship and apply deterministically.

Metrics. On the held-out rows: coverage k/P of the union of selected groups; purity k/n of the union; net coverage $(k - \frac{\tau}{1-\tau}(n - k))/P$, the Lagrangian of coverage subject to purity $\geq \tau$; share of splits with held-out purity $\geq \tau$; description stability (mean pairwise Jaccard of the selected row sets across splits); conditions actually spent; selection time. Paired differences to **vsf** with the Nadeau–Bengio corrected variance [Nadeau and Bengio, 2003]; a mark needs $|t| \geq 2$.

Why the disjoint baseline. In the external runs the held-out purity of the **rules** union reached τ in 53 % of splits (observed rule) and 87 % (certified), against 73 % and 92 % for **vsf**, **vsf_partial** and **rules_disjoint**. A comparison of coverage alone between an overlapping and a disjoint method therefore compares different constraints.

6 Data

Twenty tables with pinned SHA-256 hashes and recorded preprocessing (Table 2): the UCI classics of subgroup discovery (**mushroom**, **nursery**, **car**, **chess**), **titanic**, three small clinical tables, two wide PMLB tables (**connect_4**, $67\,557 \times 42$; **splice**, $3\,188 \times 60$, where every method is capped at $d \leq 3$ because the size-4 families are too large for all of them alike), seven real tables from public mirrors with N from 1 470 to 48 842 and base rates from 8 to 28 % (**adult**, **hmda**, **attrition**, **credit**, **resume**, two churn tables), and two planted synthetic designs: *aligned* (one 4-cell and one 2-cell in the same four columns) and *misaligned* (two 2-condition rules on disjoint column pairs, which no full grid of four columns can show with fewer than $9 + 9$ cells). Three high-dimensional expression tables ($325\text{--}7\,070$ columns) are in the repository but excluded: exhaustive search over $d \leq 4$ needs a feature pre-filter, which would enter the family and break Theorem 1.

7 Results

7.1 The price of one grid, and where it comes from

Table 3 gives the counts over the 21 observed-purity and 9 certified configurations that are informative (some method above 1 % held-out coverage); Figure 2 shows four of them against the budget. With the observed-purity rule the full grid loses to free disjoint rules in 45 of 109 cells and wins 2; with partial centres the same schema family loses 21 and wins 5, and closes a median 85 % of the gap. Under the certificate the full grid loses 17 of 48 cells; partial centres lose 6 (all on **connect_4**, Section 7.5) and win 2. The observed-rule losses are at $B \geq 16$ (20 of 21) on the tables with the most columns to spare (**connect_4**, **mlc_churn**, **credit**, **attrition**,

Table 2: Datasets and settings (generated by `experiments/paper_tables.py` from the manifest and the configuration list).

dataset	N	M	positive value	base rate	τ / m / repeats
titanic	891	8	survived	38 %	0.7/1/5; 0.9/1/5; 0.9/20/5
breast_cancer	286	9	recurrence-events	30 %	0.7/1/5
thyroid_recurrence	383	16	Yes	28 %	0.9/1/5
breast_cancer_wisconsin	699	9	malignant	34 %	0.9/1/5
car_evaluation	1728	6	acc	22 %	0.9/1/5
nursery	12960	8	spec_prior	31 %	0.7/1/5; 0.9/1/5
mushroom	8124	21	poisonous	48 %	0.9/1/2
chess_krkp	3196	36	won	52 %	0.9/1/1
adult	48842	12	>50K	24 %	0.7/20/2
hmda	2380	12	yes	12 %	0.5/20/5
attrition	1470	19	Yes	16 %	0.5/20/5
credit	4454	13	bad	28 %	0.7/20/5
resume	4870	24	yes	8 %	0.3/20/5
wa_churn	7043	19	Yes	27 %	0.7/20/5
mlc_churn	5000	11	yes	14 %	0.7/20/5
connect_4	67557	42	loss	25 %	0.7/20/1; 0.9/20/1
splice	3188	60	site_b	24 %	0.9/20/1

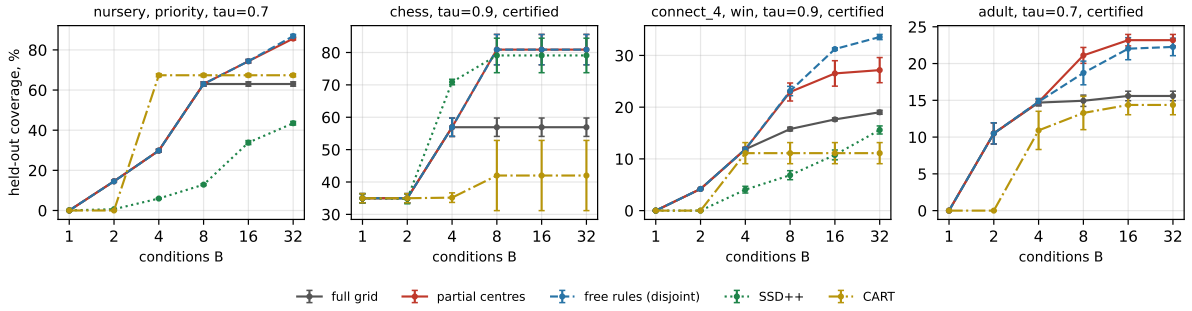


Figure 2: Held-out coverage against the condition budget on four tables (mean $\pm$ Nadeau–Bengio SE over splits). Partial centres follow free disjoint rules where the full grid stalls; **connect_4** is the table where one schema cannot hold what forty-two columns offer.

resume, **wa_churn**: Table 4), where free rules keep adding disjoint fragments from other columns after the schema’s four are exhausted.

On the misaligned design the full grid covers 0.375 at every budget (one rule; the second needs nine 4-condition cells) while partial centres reach 0.718 at $B = 16$, equal to free rules; on the aligned design the full grid still loses (0.504 vs 0.544) because the planted 2-cell costs nine full cells. At $B = 32$ the full grid spends on average 7 conditions: one grid holds few pure fully specified cells. This is the mechanism: the alignment price is the price of fixing every axis.

7.2 Under the certificate

Table 5. Only 9 of 25 certified configurations certify anything at all (the clinical tables with $N \leq 700$, **car**, **credit**, **resume**, **wa_churn**, **attrition**, **adult** at $\tau = 0.7$ and **splice** certify nothing at their τ ; Section 4). Where something is certified, one schema with partial centres ties free disjoint rules in 40 of 48 cells, wins 2 (**mushroom** at $B \leq 4$) and loses 6, all on **connect_4**; the independent check on ten other tables gave 77 ties, 2 wins, 0 losses. Held-out union purity reaches τ in 97 % of splits for every method under the certificate, against 61 % for overlapping rules under the observed rule (Table 3).

Table 3: Paired comparisons over (configuration, budget) cells, $|t| \geq 2$ with the Nadeau–Bengio correction; uninformative cells (no method above 1 % held-out coverage) left out. “Gap closed” is $(\text{cov}^P - \text{cov})/(\text{cov}_{\text{rules}_{\text{disj}}} - \text{cov})$ over cells where disjoint rules beat the grid by at least one point.

selection	comparison	better	tie	worse
certified	full grid vs disjoint rules	2	29	17
certified	partial centres vs disjoint rules	2	40	6
certified	partial centres vs full grid	15	33	0
certified	gap closed by partial centres: median 1.00, mean 0.78 ($n = 18$)			
certified	held-out union purity $\geq \tau$: grid 97 %, partial 97 %, rules _{disj} 97 %, rules 97 %			
purity	full grid vs disjoint rules	2	62	45
purity	partial centres vs disjoint rules	5	83	21
purity	partial centres vs full grid	41	68	0
purity	gap closed by partial centres: median 0.85, mean 0.75 ($n = 55$)			
purity	held-out union purity $\geq \tau$: grid 79 %, partial 79 %, rules _{disj} 77 %, rules 60 %			

7.3 Net coverage and stability

Coverage alone rewards a method that adds fragments whose purity does not hold up. Net coverage, $(k - \frac{\tau}{1-\tau}(n - k))/P$ over the held-out union, is the Lagrangian of “cover the most target rows subject to purity $\geq \tau$ ”: a group at purity exactly τ adds nothing, a union below τ scores negative. At $B = 32$ the ordering reverses between the two selection rules. Under the certificate, mean net coverage over the informative cells is 0.25 for partial centres, 0.25 for the full grid, 0.24 for disjoint and for overlapping rules and 0.20 for the tree. Under the observed-purity rule it is 0.16 for the full grid and for partial centres, 0.13 for disjoint rules and the tree and 0.05 for overlapping rules; on net coverage partial centres beat disjoint rules in 44 of 109 cells and lose 2: the freer the description, the more of its training coverage is fragments that do not survive the held-out fold (**titanic**, $\tau = 0.7$: overlapping rules -0.24 , disjoint rules 0.07 , partial centres 0.08 , full grid 0.15). With $m = 20$ the three disjoint methods are within a few points of each other on every table. The lesson is a selection-discipline effect, not a grid effect: the full grid’s cells are large because they are few, and a size floor gives the freer methods the same protection.

Description stability (mean pairwise Jaccard of the selected rows across splits) does not separate the methods: medians 0.82–0.92 under both rules, the tree highest (0.90–0.92) and partial centres and disjoint rules lowest (0.82–0.87). A freer description is chosen from more candidates and varies more between samples; the difference is small.

7.4 Trees and greedy chains

A depth- ≤ 4 CART tree with `min_samples_leaf = m`, the best over criterion and class weight at each budget, is the natural single-picture competitor. Under the observed-purity rule partial centres beat it (paired $|t| \geq 2$) in 39 of 109 cells and lose 10 (at $B = 32$: 9 wins, 12 ties, no loss); under the certificate 18 / 16 / 14 (at $B = 32$: 4 / 3 / 2), but the tree’s filter is not a certificate: its leaves are grown on the target, not drawn from a family fixed before the target is read, so the 51.7 vs 34.2 it scores on **titanic** at $\tau = 0.7$ carries no guarantee (its held-out purity there is 89 %, below τ at the certified level). On the planted grid design CART is greedy at the root and covers 2–4 % (`experiments/compare_baselines.py --synthetic grid`). An optimal-tree solver with the coverage-under-purity objective is the right competitor and is future work.

The greedy schema chain $S_1 \subset S_2 \subset S_3 \subset S_4$ (add the column that most improves the ranking key) is worse than the exhaustive winner in 33 of 109 cells under the observed rule and 9 of 48 under the certificate, never better (mean gap at $B = 32$: 6.2 and 3.6 points); on **chess**

Table 4: Held-out coverage (%) at $B = 32$, observed-purity selection. Marks: partial centres vs the full grid (p vs g) and vs disjoint rules (p vs r).

dataset	τ	grid	partial	rules _{disj}	rules	tree	p vs g	p vs r
titanic	0.9	48	53	57	59	51	+	-
titanic	0.7	67	73	77	85	70	+	-
titanic	0.9	47	46	47	53	47	=	=
breast_cancer	0.7	27	32	39	49	30	=	=
thyroid_recurrence	0.9	83	88	88	89	85	=	=
breast_cancer_wisconsin	0.9	82	85	75	94	89	=	+
car_evaluation	0.9	21	21	12	12	0	=	+
nursery	0.9	42	57	59	68	28	+	-
nursery	0.7	63	86	87	90	67	+	-
mushroom	0.9	99	100	100	100	100	+	=
chess_krpk	0.9	84	84	92	98	42	=	-
adult	0.7	31	42	44	63	20	+	=
hmda	0.5	19	29	27	51	30	+	=
attrition	0.5	34	43	47	69	34	+	-
credit	0.7	17	24	30	50	17	+	-
resume	0.3	4	4	15	18	4	=	-
wa_churn	0.7	30	33	37	49	30	+	-
mlc_churn	0.7	16	25	33	50	21	+	-
connect_4	0.7	10	10	19	27	11	+	-
connect_4	0.9	27	30	39	55	11	+	-
splice	0.9	0	0	0	0	0	=	=

it never leaves the 1D winner (35 % against 84 % and 57 % exhaustive). This is the synergy case the exhaustive search exists for.

7.5 SSD++

SSD++ optimises compression of the whole target column, not coverage at a purity floor, and the comparison shows both sides of that. On held-out coverage at τ partial centres beat it in 63 of 109 observed-purity cells and lose 15 (certified: 29 / 16 / 3); at $B = 32$ its mean coverage is 0.40 against 0.46 (certified 0.35 against 0.42). On held-out purity it is the best method in the study: 0.96 under the certificate and 0.83 under the observed rule against 0.91 and 0.76 for partial centres, and its net coverage is highest under both rules (0.33 and 0.27 against 0.28 and 0.14). The two facts are the same fact: a rule list stops adding rules when they no longer pay for themselves in code length, so what it returns is small and clean; a coverage objective keeps adding groups at exactly τ . Which is wanted depends on the question, and the paper’s task fixes it at coverage. One more property of the list matters for the comparison: its i -th rule applies to the residue of the first $i - 1$, including rules for the *other* class, so a later rule can be pure on the residue where no conjunction is pure on the table. On **splice** ($\tau = 0.9$, $m = 20$) every conjunctive family of length ≤ 3 is empty on every training fold, all grid and rule methods cover 0.4 %, and SSD++ covers 20 % at held-out purity 0.81, below τ . Ordered “else” semantics is a description language of its own; partial centres could adopt it (a plate minus an earlier bar is still a region of the same cube) and we leave that as future work.

Where the grid still costs: connect_4. All six certified losses to disjoint rules are on **connect_4** (67557×42 , three values per cell): at $\tau = 0.9$, $B = 32$, partial centres cover 27 % against 34 % for disjoint rules (43 % overlapping); at $\tau = 0.7$, 6 % against 12 %. Free rules choose their four columns per rule and forty-two columns give them many pure four-cell conjunctions on different boards; one schema cannot hold them. This is the case the one-grid constraint was

Table 5: Held-out coverage (%) at $B = 32$, certified selection (configurations where some method certifies anything).

dataset	τ	grid	partial	rules _{disj}	rules	tree	p vs g	p vs r
titanic	0.7	34	34	34	35	52	=	=
nursery	0.9	42	49	49	58	23	+	=
nursery	0.7	40	61	67	68	67	+	-
mushroom	0.9	96	98	98	100	96	+	=
chess_krkp	0.9	57	81	81	94	42	+	=
hmda	0.5	15	15	15	15	15	=	=
mlc_churn	0.7	9	9	9	9	12	=	=
connect_4	0.7	6	6	12	17	10	=	-
connect_4	0.9	19	27	34	43	11	+	-

expected to lose, and it does; on **chess** ($M = 36$) partial centres and disjoint rules tie at 81 % where the full grid has 57 %.

7.6 Ablations: capacity rule and size floor

`experiments/ablations.py` varies the capacity divisor ($N_{\text{train}}/5$, $/10$, $/20$, no merging) and the size floor $m \in \{1, 5, 20\}$ on eight tables at $B = 32$. The capacity rule is inert on seven of them: the winning schemas never reach the ceiling, and coverage changes by at most 0.1 points. On **car_evaluation**, a full factorial design of 1728 rows at the ceiling, every change loses coverage — -7.1 points with a looser rule ($N/5$) or none, -8.6 with a tighter one ($N/20$) — for the full grid and for partial centres alike: merged levels pool sparse cells above the floor, and merging further splits the target. The rule is therefore not a tuning knob that favours the method; it is a display constraint that costs nothing on most tables and a few points on one.

The size floor is the lever that matters, and it acts on the freer descriptions most. From $m = 1$ to $m = 20$ disjoint rules lose 4.6 points of coverage and raise the share of splits whose held-out purity reaches τ from 77 to 91 %; partial centres lose 3.7 points (85 $\rightarrow$ 87 %); the full grid loses 3.0 (94 $\rightarrow$ 83 %: its cells being large already, the floor removes the few small pure ones it had). Conditions spent at $B = 32$ fall from 29 to 14 for rules and from 25 to 13 for partial centres. Section 7.3 is the same effect seen through net coverage.

7.7 Cost

Selection time per split (mean over budgets and selection rules, one core): full grid 7.3s and partial centres 8.4s on **adult** ($N = 48\,842$, $M = 12$); 5.9 / 9.8s on **chess** ($N = 3\,196$, $M = 36$, 58 905 schemas of size 4); 2.2 / 9.6s on **mushroom**; under one second on every table with $M \leq 20$ and $N \leq 8\,000$. The partial search enumerates no new cells; its extra cost is the disjoint packing over the pool of each schema, which grows with the number of qualifying sub-cells (**mushroom**, $\tau = 0.9$, has many). Free rules cost the same as the grid on every table; the tree is fastest.

8 Conclusion

The alignment constraint of a one-grid display was measured, not assumed. Against free disjoint rules of the same length its price with fully specified cells is real (the full grid loses 45 of 109 observed-purity cells and 17 of 48 certified ones); the price is the requirement to fix every axis, not the requirement to share columns, and partial-axis centres remove it on every table but the widest (**connect_4**, where one schema of four columns cannot hold what forty-two columns offer to free rules) while keeping the disjointness, the certificate and the picture. The certificate itself is a family Bonferroni over row sets with a threshold valid for heterogeneous rows; it survives

the search by construction, costs logarithmically in the number of columns and heavily in purity, and is empty on small tables. The comparison also fixed two things about how such methods should be compared: an overlapping union of rules each at purity τ is not at purity τ , and a size floor is what keeps free descriptions honest on held-out data. Two directions follow. Ordered (“else”) semantics, which lets SSD++ find pure residues on `splice` where no conjunction is pure, can be given to partial centres without leaving the cube. And whether the shared picture buys readability is a question about people, not tables; it needs a user study and is not claimed here.

9 Limitations

The relaxation makes the aligned method closer to free rules; what remains of the constraint is justified by the shared picture, and we have not measured readability. The certificate’s power at $N \leq 500$ is negligible; with $\tau = 0.9$ it certifies nothing on `titanic`. The capacity rule $\lfloor N/10 \rfloor$ interacts with the folds (`car`: a full factorial at the capacity threshold is merged in the fold but not on all rows, +15 points of apparent gap). Numeric columns are binned before the search; the bins are not optimised. Seventeen tables with $M \leq 60$ is the scope of the evidence; `splice` shows that at $\tau = 0.9$ with $m = 20$ every conjunctive family of length ≤ 3 can be empty, and `connect.4` that the one-grid constraint still costs six to seven points where the columns are many and alike. High-dimensional tables need a target-free feature pre-filter before the family is fixed; we did not build one.

10 Reproducibility

Code: [\[Zenodo DOI\]](#). Every number is produced by a script in `experiments/` with seed 0 from data loaded through the manifest. Use of AI assistance: [\[disclosure per journal policy\]](#).

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